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CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
Acceleration due to gravity	g	=	9.80 m s^{-2}
1 Atmosphere	Atm	=	1.00×10^5 N m^{-2}
Boltzmann's constant	k	=	1.38×10^{-23} J K^{-1}
Density of water		=	1.00×10^3 kg m^{-3}
Specific heat capacity of water		=	4200 J kg^{-1} K^{-1}
Specific latent heat of fusion of ice		=	3.34×10^5 J kg^{-1}
Specific latent heat of vaporization of water		=	2.26×10^6 J kg^{-1}
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Stefan-Boltzmann constant	σ	=	5.67×10^{-8} W m^{-2} K^{-4}
Speed of light in free space	c	=	3.00×10^8 m s^{-1}

SECTION A

Attempt ALL questions. You MUST write in this answer booklet.

1. A student is set the task of determining the density of the metal used to make ball bearings.

(a) To find the diameter of the bearings the student lined up 10 identical balls in a row and placed a ruler against them as shown in Figure 1.

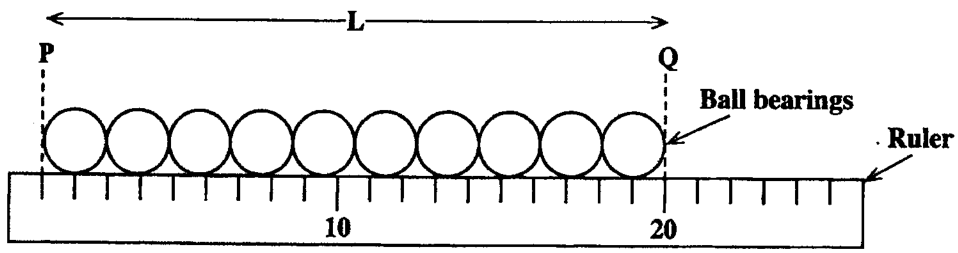

Figure 1

The positions on the scale were estimated to be: P 1.0 ± 0.2 cm, Q 20.2 ± 0.2 cm

(i) Write down the value for the distance L using the same format

$L =$ ± cm **[1 mark]**

(ii) What value does this imply for the diameter D of one of the spheres?

$D =$ ± cm **[1 mark]**

(b) The only balance available to the student was a 0 - 500 g instrument whose smallest division is 5 g. Using this balance the student obtained a mass of 30.4 ± 0.3 g for one of the ball bearings.

Suggest how the mass of one of the ball bearings could have been determined to the precision indicated above.

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[3 marks]

(c) Find the density of the metal used to make the ball bearings, giving your answer to a suitable number of significant figures together with the error (uncertainty) in the value.

(Volume of a sphere = $\frac{1}{6}\pi D^3$) **[5 marks]**

Total 10 marks

2. When you blow across the top of a soft-drink bottle as shown in Figure 2, a tone is heard. The pitch of the sound depends on how much drink is left in the bottle. This phenomenon is called cavity resonance.

Scientists investigating cavity resonance have deduced a formula for the frequency produced:

$$f = \frac{c}{2\pi}\sqrt{\frac{A}{LV}}$$

where c = velocity of sound in the cavity
A = area of the neck of the bottle
L = length of the neck of the bottle
V = volume of air in the cavity

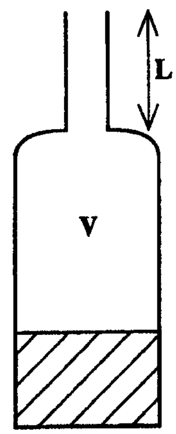

Figure 2

Using a small speaker connected to a signal generator the resonant frequency may be found for various values of the air volume. The data obtained from one such experiment are shown in Table 1. The bottle used had a neck length of 5.9×10^{-2} m and the cross-sectional area of the neck was 2.55×10^{-4} m^2.

Table 1

V/m^3	$1/\sqrt{V}$	f/Hz
250×10^{-6}		225
200×10^{-6}		255
150×10^{-6}		290
125×10^{-6}		320
100×10^{-6}		350
80×10^{-6}		400
65×10^{-6}		445

(a) Complete Table 1 and then plot a graph of f against $1/\sqrt{V}$ using the grid opposite. There is no need to start the graph at the origin. **[5 marks]**

GO ON TO THE NEXT PAGE

(b) Find the gradient of the graph.

[2 marks]

(c) Use your answer to (b) to determine the velocity of sound in the bottle.

[3 marks]

Total 10 marks

3. The data shown in Table 2 was collected by hanging weights (F) on a metal wire and measuring the extension (x) caused. The wire was originally 2.00 m long and had a cross-sectional area of 2.00×10^{-7} m^2.

Table 2

F/N	x/mm
5	0.54
10	1.08
15	1.61
20	2.15
25	2.84
30	3.95
35	5.38
40	8.00

(a) Plot a graph of the weight, F, against the extension, x, on the graph grid opposite.

[4 marks]

(b) By estimating the area under the graph, find the work done in stretching the wire by 4.0 mm.

[2 marks]

(c) (i) At what load does the wire cease to obey Hooke's law?

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[1 mark]

(ii) Use the graph to determine the value of Young's Modulus for this metal.

[3 marks]

Total 10 marks

SECTION B

You must attempt THREE questions from this section. Choose ONE question EACH from Module 1, 2 and 3. You MUST write your answers in the separate answer booklet provided.

MODULE 1

Answer EITHER Question 4 OR Question 5.

4. (a) (i) Distinguish between scalar and vector quantities and give ONE example of EACH.

(ii) Three forces are acting at a point O as shown in Figure 3. Find their resultant and clearly indicate the direction of this resultant force with respect to the positive x-axis.

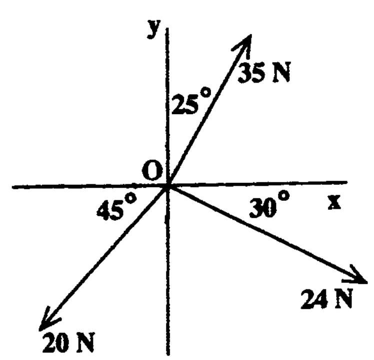
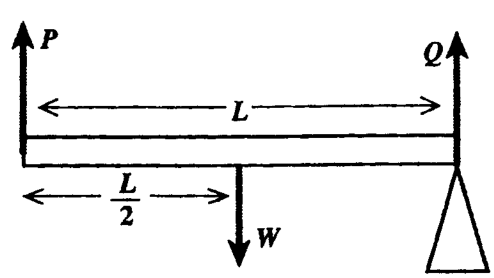


Figure 3

[10 marks]

(b) (i) An extended object (e.g. a uniform wooden plank) is acted upon by several forces. State TWO conditions for the object to remain in equilibrium. Using these two conditions, write TWO equations relating the forces and distances for the situation shown in Figure 4.



(ii) Figure 5 depicts a uniform plank, weight 200 N, leaning against a smooth wall. The foot of the plank is 3 m from the wall and the top of the ladder is 6 m from the ground.

Find

a) The upward push, *R*, of the ground on the plank.

b) The magnitude of the normal force, *P*, exerted by the wall on the top of the plank.

c) The magnitude and direction of the resultant of *R* and *F*, the friction force at the ground.

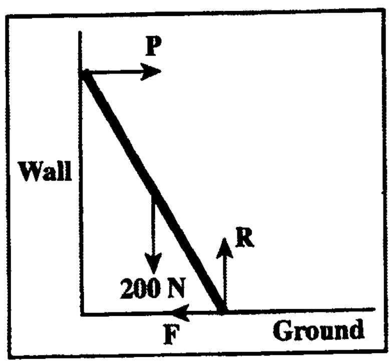


Figure 5

[10 marks]

Total 20 marks

5. (a) (i) A body of mass m is accelerated from rest to a speed v by the application of a constant force. Derive the equation $E_k = \frac{1}{2}mv^2$ for the kinetic energy of the body.

(ii) A satellite is launched from the ground and placed in an orbit with a height of 900 km. Why does the equation $\Delta E_p = mgh$ NOT apply to this situation?

(iii) The car on a roller coaster (Figure 6) is travelling at 12 m s^{-1} at the top of a hill but a short time later when it reaches the bottom of the hill, 35 m below, it is travelling much faster.

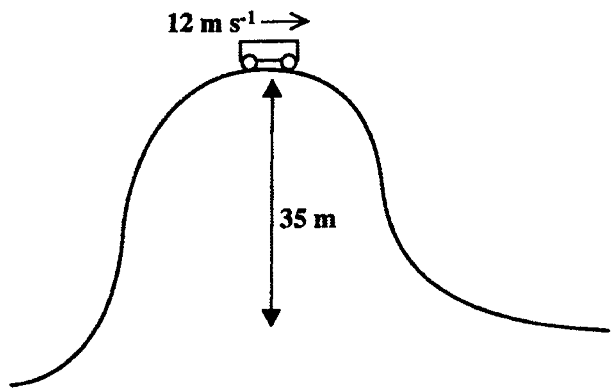


Figure 6

Assuming frictional losses are negligible, write an equation to represent the energy transformation occurring and use it to find the speed of the car at the bottom of the hill.

[9 marks]

(b) On a construction site a winch is used to raise a paint filled drum, weighing 1200 N, through a height of 35 m. It takes 24 s to lift the drum.

(i) What was the gain in potential energy of the load?

(ii) Find the power output of the winch motor.

(iii) The winch has an efficiency of 70%. How much electrical power is required by the winch's motor? **[6 marks]**

(c) While the paint drum in (b) is stationary at a height of 35 m the cable breaks.

(i) Sketch graphs to show the variation with time of the drum's displacement and velocity.

(ii) Find the time taken by the drum to reach the ground.

MODULE 2

Answer EITHER Question 6 OR Question 7.

(a) Figure 7 shows a section of a stationary transverse wave.

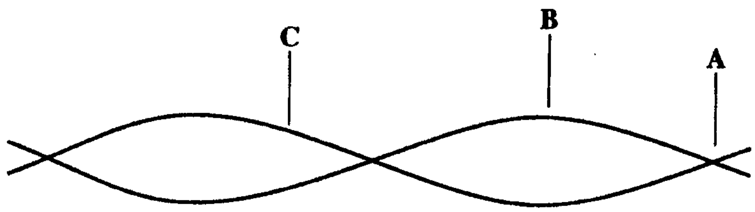


Figure 7

(i) Explain how such a wave may be formed from two progressive waves, CLEARLY indicating any conditions that MUST be fulfilled.

(ii) Compare the amplitudes of particle vibrations at A, B and C and the phases at B and C.

(iii) Give the THREE subjective sensations of sound and state what objective quantities are associated with them. **[8 marks]**

(b) A loudspeaker at a concert generates 1×10^{-2} Wm^{-2} at a distance of 20 m from the speaker at a frequency of 1 kHz.

(i) Calculate the wavelength of the waves emitted by the loudspeaker.

(ii) What is the intensity level at 20 m?

(iii) Given that the intensity (in W m^{-2}) of the sound is inversely proportional to the distance, x, from the speaker ($I \; \alpha \frac{1}{x^2}$), find the distance at which the intensity level will be at the pain threshold of 120 dB.

[Threshold intensity of hearing = 1×10^{-12} W m^{-2}
Velocity of sound = 340 ms^{-1}] **[12 marks]**

Total 20 marks

7. (a) (i) Define the term 'refractive index' of a medium as it applies to light waves.

(ii) Draw a diagram showing the refraction of waves moving from one medium to another in which the velocity is slower.

(iii) Derive the law of refraction, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, for waves moving from one medium to another.

(iv) State the relationship between the frequencies of the waves in the two media and the relationship between wavelengths in the two media. **[8 marks]**

(b) The refractive index of a glass prism is 1.51 for red light and 1.55 for blue light.

(i) What is the speed of EACH of these colours of light in the glass?

(ii) A mixture of red and blue light is incident on the prism as shown in Figure 8. What is the angle between the two emergent rays? **[6 marks]**

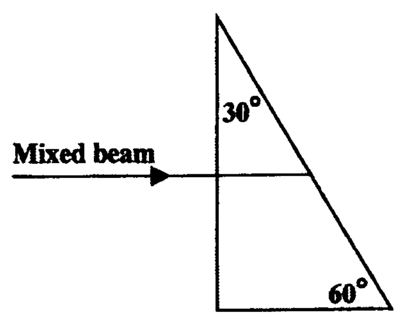


Figure 8

(c) A larger dispersion of the two colours of light may be produced by using a diffraction grating. What is the angle between rays of the two colours if a grating with 1200 lines per millimetre is used? The wavelength of the red light is 678 nm and that of the blue light is 422 nm. **[6 marks]**

Total 20 marks

MODULE 3

Answer EITHER Question 8 OR Question 9.

(a) (i) State ONE situation in which a constant volume gas thermometer would be used in preference to other types of thermometers. Give a reason for your choice.

(ii) State, with a reason, the type of thermometer you would use to

a) locate the hottest part of a Bunsen burner flame

b) record the mean temperature of the gases emerging from the exhaust pipe of a car.

(iii) State TWO ways in which the behaviour of a thermistor differs from that of a platinum coil when used as a resistance thermometer.

[8 marks]

(b) The resistance of a thermistor is given by the formula:

$R = R_o e^{B/T}$, where R is in ohms (Ω), T is in kelvins, and R_o and B are constants that can be determined by measuring R at calibration points such as the ice point and the steam point.

(i) If R = 7360 Ω at the ice point and 153 Ω at the steam point, calculate the values of R_o and B.

(ii) Determine the temperature, in degrees Celsius, when the thermistor resistance is 2200 Ω.

(iii) a) If an empirical temperature scale were used, what temperature would this resistance of 2200 Ω give?

b) Explain why this scale is NOT suitable for determining the value of the temperature from resistance.

[12 marks]

Total 20 marks

9. (a) (i) Use the kinetic theory of gases to explain

a) how the molecules inside a gas cylinder containing oxygen are able to exert a pressure

b) why pumping in more oxygen increases the pressure inside it

c) why the pressure would increase if the cylinder were left in the sun for a long time.

(ii) A cylinder contains 2.8 mol of oxygen at room temperature, 27°C, and the pressure inside the cylinder is 4.5×10^5 Pa. The temperature rises to 47 °C when another 3.9 mol of oxygen is pumped into the cylinder. Find the new pressure inside the cylinder.

[10 marks]

(b) (i) Write an equation representing the first law of thermodynamics and state CLEARLY the meaning of EACH term.

(ii) The molar heat capacity at constant volume, C_V , of a gas is the quantity of thermal energy required to raise the temperature of one mole of the gas by one kelvin without any change in volume.

a) How much thermal energy would be required to raise the temperature of 6.2 mol of a gas from 25°C to 50°C while the volume remained constant if $C_V = 12.5\ J\ K^{-1}\ mol^{-1}$?

b) By HOW MUCH would the internal energy of the gas change?

c) If instead, the gas were heated at constant pressure from 25°C to 50°C the thermal energy required would be 3200 J. How much work would be done by the gas in this case?

d) Deduce the value of C_p , the molar heat capacity of the gas, at constant pressure.

[10 marks]

Total 20 marks